Factory Resume

Operations recovery and capacity optimisation for disrupted e-commerce fulfilment — product case

The problem

A disruption event — illness, attrition, a facility issue — removes a large share of a fulfilment centre's workforce overnight. Orders, inventory, machines, and logistics partners are all still there, but there isn't enough staff to run the operation at normal throughput. Within hours, an operations manager must decide which orders still ship on time, how to redeploy the remaining workers, how much work to push to external partners, and which customers will be told their order is delayed — and defend those calls to leadership. Today that decision gets made under pressure, in spreadsheets, with no shared view of the actual constraint.

Target users and their pain points

Primary user: the operations / fulfilment manager at a mid-size D2C e-commerce brand, accountable for on-time dispatch during normal operations and the first person leadership calls during a disruption.

- No single view of the disruption's real impact — capacity, inventory, and order risk live in separate spreadsheets and tribal knowledge.
- Staffing reallocation is done by gut feel; it's easy to accidentally assign someone to a task they aren't qualified for, or to starve a safety-critical step like quality inspection.
- Outsourcing decisions are made without clear cost-per-unit or capacity math, so partners get over- or under-committed.
- There's no fast way to test “what if another 20% of staff are out tomorrow?” before it actually happens — every what-if is a manual re-plan.

The solution: Factory Resume

Factory Resume is a decision-support tool that turns a disrupted operation's raw data — orders, staff, tasks, inventory, and external providers — into one feasible recovery plan, and lets the manager stress-test it against further disruption before committing. It is built decision-engine first: every number on screen comes from a deterministic calculation over the current data, not a static mock-up, so the plan updates the moment reality changes.

How workforce allocation, outsourcing, and recovery planning work

Workforce allocation	For every task, required labour hours come from order demand; available staff are counted only if they hold the right skill *and* are eligible for that specific task. Safety- and quality-critical tasks are staffed first regardless of demand; everything else is ranked by a transparent, demand-weighted priority score.

Outsourcing	Wherever internal capacity still falls short, eligible external providers are matched by cost, location, and activation time — never assuming a partner can take unlimited volume. Every recommendation states the gap it closes, its cost, and its risks (quality terms, slow activation, partial coverage).
Recovery planning	Internal allocation and outsourcing are combined with order priority and usable inventory to resolve, order by order, what actually ships, what's delayed, and why (capacity or stock). Scenario controls apply a further disruption (staff loss, provider outage, demand spike, material delay) and instantly show the new plan against the baseline.

Core user flow and MVP features

Overview (state of the operation, biggest bottleneck) → **Workforce** (required vs. available staffing, allocation order) → **Outsourcing** (recommended engagements, cost, risk) → **Recovery Plan** (full order-by-order outcome, manager approval) → **Scenarios** (simulate a further disruption, compare against baseline).

MVP deliberately excludes: live marketplace/WMS/logistics integrations, autonomous booking of partners, real payroll or employee scheduling, and a custom-trained model — the value is proven with simulated data and a deterministic engine before any of that is warranted.

Success metrics and why this is valuable

- **% of demand fulfilled under disruption** and **orders at risk** — the plan should visibly protect high-priority orders first; in the current demo scenario the baseline plan lifts fulfilment from an unmanaged ~13% to 56% by correctly staffing the one bottleneck task and recommending the right outsourcing.
- **Time to a defensible recovery plan** — minutes instead of a multi-hour manual spreadsheet exercise, with every number traceable to an input.
- **Incremental outsourcing spend vs. units recovered** — cost-per-unit-saved, visible before committing budget.
- **Scenario coverage** — how many plausible further disruptions were tested before the real one happened.

The value isn't a prettier dashboard — it's replacing an ad hoc, undocumented scramble with a repeatable, explainable decision a manager can stand behind, and that leadership can audit after the fact.